



Department of Computer Science

CS3006 – Parallel and Distributed Computing

FALL 2025 (BCS-6E, BCS-6F)

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Office Location/Number: F-36 (New Building)	Office Location/Number: N/A
Office Hours: Mon & Thu 14:00 - 15:00	Office Hours: available upon request

Course Information

Program:	BS	Credit Hours:	3
Type:	Core	Pre-requisites:	CS220
Website: BCS-6E	https://classroom.google.com/c/Njg3OTE2Nzk3Mjc5		
Website: BCS-6F	https://classroom.google.com/c/NzQ1ODQ3ODYwMzM0		
Class Meeting Time:	Mon and Wed (8:30am-10:00am BCS-6F); (10:00am-11:30am BCS-6E)		
Class Venue:	NB-209		

Course Description/Objectives/Goals:

This course covers a broad range of topics related to parallel and distributed computing, including parallel and distributed architectures and programming paradigms of parallel and distributed systems. Basic goal of this course is to understand the fundamental concepts of parallel and distributed computing, analyze different problems and develop programming solutions of parallel problems.

Initially we will discuss an introduction to parallel and distributed systems and the categorization of multiprocessor systems according to Flynn's Taxonomy. Overall the course is divided into two parts. The first part covers parallel shared memory systems and programming shared memory machines with P-threads and OpenMP. The second part covers distributed systems, distributed system architectures, types of distributed systems (Clusters, Grid and Cloud computing), fault tolerance techniques and finally programming distributed systems using MPI.

Program Learning Outcomes (PLOs):

This course covers the following PLOs:

PLO#	PLO Name	PLO Description
PLO 2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
PLO 3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using

		fundamental principles of mathematics, computing sciences, and relevant domain disciplines
PLO 4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

Course Learning Outcomes (CLOs):

At the end of the course students will be able to:

CLO #	CLO description	BT domain / BT Level	PLO #
CLO 1	Demonstrate understanding of various concepts involved in parallel and distributed computer architectures.	C3 (Applying)	PLO 2
CLO 2	Implement different parallel and distributed programming paradigms and algorithms using Message-Passing Interface (MPI) and OpenMP.	C3 (Applying)	PLO 4
CLO 3	Perform analytical modelling, dependence, and performance analysis of parallel algorithms and programs.	C4 (Analyzing)	PLO 3

Course Textbook

- **[AAGV]** Introduction to Parallel Computing, 2nd Edition by Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar
- **[KLEP]** Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems, by Martin Kleppmann, ISBN-13 : 978-1449373320
- Using OpenMP: Portable Shared Memory Parallel Programming by Barbara Chapman, Gabriele Jost, Ruud van der Pas.

Reference books

- **[MA]** Distributed Systems, 4th edition (version 4.02, dated Feb 2024) by Maarten Van Steen and Andrew Tanenbaum (Freely available at: <https://www.distributed-systems.net/index.php/books/ds4/ds4-ebook/>)
- Distributed and Cloud Computing: Clusters, Grids, Clouds, and the Future Internet, K Hwang, J Dongarra and GC. C. Fox, Elsevier, 1st Ed.
- Distributed Systems: Concept and Design by George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair, 5th edition

Tentative Lecture Schedule

Lecture	Topics to be covered
1	Course Introduction Introduction to parallel and distributed systems, Motivating parallelism (evolution, need, and future) [AAGV – Chapter 1]
2	The scope of parallel computing in commercial, scientific, and engineering and design applications, Scalability issues, Flynn's Taxonomy [AAGV – Chapter 2 and 5]
3	Amdahl's law, Karp-Flatt Metric, Gustafson's law, Processor to memory connection strategies
4	Network topologies for parallel architectures; Evaluating static inter-connections in-terms of diameter, arc-connectivity, bisection width, and cost. Cut-through Routing and Cost-

	performance tradeoffs [AAGV – Chapter 2 and 5]
5	Principles of parallel algorithm design, Dependency Graphs, Granularity, Concurrency [AAGV – Chapter 3]
6	Decomposition Techniques, Task Interaction Graph, Mapping Techniques [AAGV – Chapter 2 and 5]
7	Programming Shared Address Space Platforms using POSIX Thread API and OpenMP (Thread Basics, Motivation, Synchronization Primitives)
8	Shared memory programming with OpenMP
9	Parallel programming with OpenMP, work sharing constructs, Synchronization Constructs in OpenMP, OpenMP Library Functions, Environment variables
10	Revision
11	Introduction to Distributed Systems, Types of Distributed System Architectures such as clusters, grids, cloud and utility computing
12	Distributed Operating Systems Case study 1: Kubernetes Scheduling: Taxonomy, Ongoing Issues and Challenges (https://dl.acm.org/doi/pdf/10.1145/3539606) Case study 2: Large-scale Cluster Management at Google with Borg (https://dl.acm.org/doi/pdf/10.1145/2741948.2741964)
13	Wrapping-up case studies from last lecture Chapter 8 of [KLEP] : The trouble with distributed systems
14	Chapter 8 of [KLEP] : The trouble with distributed systems
15	Basic Communication Operations (Broadcast, Reduction, Scatter, Gather and Circular Shift) [AAGV – Chapter 4]
16	Parallel Cost analysis for the operations over Ring, 2D-Mesh, Hypercube, and 3D-cube [AAGV – Chapter 4]
17	Parallel Cost analysis for the operations over Hypercube and 3D-cube [AAGV – Chapter 4]
18	Programming Distributed machines using Message passing interface (MPI)
19	Collective Communication and Computation Operations: Barrier, Broadcast, reduction
20	Collective Communication and Computation Operations: prefix, Gather, scatter, All-to-All
21	Revision
22	Fault Tolerance Techniques: Hardware Redundancy, Information Redundancy, Time Redundancy
23	GRPC and Protocol Buffers (https://grpc.io/docs/what-is-grpc/introduction/)
24	MapReduce, and GFS (Hadoop and HDFS as open-source examples of the above two) MapReduce: https://static.googleusercontent.com/media/research.google.com/en//archive/mapreduce-osdi04.pdf Google File System (GFS): https://static.googleusercontent.com/media/research.google.com/en//archive/gfs-sosp2003.pdf
25	Logical Clocks, Vector Clocks. Raft Consensus https://www.cs.princeton.edu/courses/archive/fall24/cos316/lectures/L11-concurrency-time.pdf https://www.cs.princeton.edu/courses/archive/fall24/cos316/lectures/L12-vc.pdf https://www.cs.princeton.edu/courses/archive/fall24/cos316/lectures/L13-consistency.pdf
26	Raft Consensus. RAFT: election process https://raft.github.io/raft.pdf

	https://www.cs.princeton.edu/courses/archive/spring21/cos418/docs/L13-consensus-raft.pdf
27 onwards	Revisions and Project Evaluations

(Tentative) Grading Criteria

1. Assignments and Project (15%)
2. Quizzes (10%)
3. 2 Midterm Exam(s) (30%)
4. Final Exam (45%)

Grading scheme

The grading scheme followed will be **absolute** in accordance with the university standards.

Passing Criteria

Students need to score a minimum of **50%** to pass the course.

Course Policies

1. Students are expected to attend all sessions. However, they might avail 20% leaves in emergency situations. Beyond this the student will not be allowed to appear in the final exam.
2. Plagiarism is not tolerable in any of its form. Minimum penalty would be an 'F' grade in the course. Automated tools may be deployed to detect pirated copies. Students bear all the responsibility for protecting their assignments. In case of cheating, both parties will be considered equally responsible.
3. Assignments must be submitted in time. No late submissions will be accepted and/or awarded. REMEMBER that the overall submission time allowed includes the extra time given during which SLATE/Google Classroom doesn't work. Therefore, deadlines are firm.
4. Rechecking of quizzes/assignments must be done within one week of it being uploaded on FLEX. In case they are shown to you during the class, the week starts thereon.